The Korteweg–de Vries Equation

Everyone in some order

April 19, 2026

Outline

Demo TODO: Select a section!

sunday 10 am

Notation

- $u(x, t)$: wave profile
- $x \in \mathbb{R}$: spatial coordinate
- $t \geq 0$: time
- h: undisturbed water depth
- ∂_x, ∂_t: partial derivatives
- $H^s(\mathbb{R})$: Sobolev space
- $\|\cdot\|_{H^s}$: Sobolev norm

The KdV Equation

$$u_t + 6uu_x + u_{xxx} = 0$$

- Models shallow water waves
- Balance of nonlinearity and dispersion

History

- Observed by John Scott Russell (1834)
- Derived by Korteweg and de Vries (1895)
- First example of soliton behavior

Modeling Assumptions

- Shallow water: wavelength $\gg h$
- Weak nonlinearity
- Weak dispersion
- Unidirectional propagation

need to decide $(-L, L)$ or $(-\infty, \infty)$

I put some hints on higher order conserved terms in a separate doc!

does not work on $(-L,L)$ or $(-\infty,\infty)$

Plan: Local ⇒ Global Well-Posedness

Starting from $u_0 \in H^3_{\text{per}}(-L, L)$, we want a unique solution of KdV for all time.

Strategy:

- Step 1: Control $\|u\|_{H^3}$ by an energy estimate. ✓
- Step 2: Turn the estimate into a local solution (Picard iteration). ✓
- Step 3: Use conservation laws to rule out blow-up ⇒ global. ✓

Why H^3? KdV has three x-derivatives, so we need at least three derivatives of u in L^2 to write the equation classically. ✓

Step 1: The H^3 Energy Estimate

Apply ∂_x^3 to KdV:

$$(u_{xxx})_t + 6\,\partial_x^3(uu_x) + u_{xxxxxx} = 0.$$

Multiply by u_{xxx} and integrate over $(-L, L)$. By periodicity:

$$\int u_{xxx}\, u_{xxxxxx}\, dx = 0 \quad \text{(dispersive term cancels).}$$

Hardest nonlinear term reduces by parts:

$$\int u\, u_{xxxx}\, u_{xxx}\, dx = -\tfrac{1}{2}\int u_x\, u_{xxx}^2\, dx.$$

Remaining terms handled by product estimate $\|uu_x\|_{H^3} \le C\|u\|_{H^3}^2$:

$$\tfrac{d}{dt}\|u\|_{H^3}^2 \le C\,\|u\|_{H^3}^3.$$ ← not sure!

not sure how this works

Bounds are true. Something went wrong with the derivation.

Can use cons quant.

Step 2: Local Existence

Let $y(t) = \|u(t)\|_{H^3}$. Previous estimate gives $y' \leq Cy^2$. Nonlinear Gronwall:

$$\|u(t)\|_{H^3} \leq \frac{\|u_0\|_{H^3}}{1 - Ct\|u_0\|_{H^3}}, \qquad t < T_* = \frac{1}{C\|u_0\|_{H^3}}.$$

A priori bound on $[0, T_*)$.

Construct the solution via Duhamel form with Airy group $S(t)$:

$$u(t) = S(t)u_0 - 6\int_0^t S(t-\tau)(uu_x)(\tau)\,d\tau.$$

Picard iteration $\Rightarrow$ unique $u \in C([0,T]; H^3_{\text{per}})$ for small T.

Why Local Isn't Enough

The bound

$$\|u(t)\|_{H^3} \leq \frac{\|u_0\|_{H^3}}{1 - Ct\|u_0\|_{H^3}}$$

blows up as $t \to T_*$.

- Either the solution actually blows up (bad).
- Or the estimate is too weak (must prove persistence).

Key idea: if $\|u(t)\|_{H^3}$ stays bounded on every finite interval, we can restart the local theorem and continue forever.

Conservation laws give us exactly this.

Step 3a: H^1 Bound from Conservation Laws

Recall two conserved quantities:

$$I_1 = \int u^2\,dx, \qquad I_2 = \int (u_x^2 - 2u^3)\,dx.$$

Rearrange I_2:

$$\|u_x\|_{L^2}^2 = I_2 + 2\int u^3\,dx \leq I_2 + 2\|u\|_{L^\infty}\|u\|_{L^2}^2.$$

Apply Sobolev embedding $\|u\|_{L^\infty} \leq C\|u\|_{H^1}$ and Young's inequality to absorb $\frac{1}{2}\|u_x\|_{L^2}^2$ to the left:

$$\tfrac{1}{2}\|u_x\|_{L^2}^2 \leq I_2 + C(I_1).$$

Since I_1, I_2 are constants:

$$\|u(t)\|_{H^1} \leq M(u_0) \quad \text{for all } t.$$

Step 3b: How H^1 Regularity Controls H^3

Go back to the nonlinear term in the H^3 estimate. Crude bound was $C\|u\|_{H^3}^3$. Look more carefully:

$$\left|\int (\partial_x^j u)(\partial_x^k u)\,u_{xxx}\,dx\right| \leq \|\partial_x^j u\|_{L^\infty}\|\partial_x^k u\|_{L^2}\|u_{xxx}\|_{L^2}.$$

Lowest-order factor sits in L^∞ and is controlled by H^1 via Sobolev embedding:

$$\|\partial_x^j u\|_{L^\infty} \leq C\|u\|_{H^1}.$$

So one factor of $\|u\|_{H^3}$ is really $\|u\|_{H^1}$ in disguise. Improved estimate:

$$\tfrac{d}{dt}\|u\|_{H^3}^2 \leq C(1 + \|u\|_{H^1})\|u\|_{H^3}^2.$$

Global Well-Posedness

Using $\|u(t)\|_{H^1} \leq M$ from Step 3a:

$$\frac{d}{dt}\|u\|_{H^3}^2 \leq C(M)\,\|u\|_{H^3}^2.$$

Linear (not quadratic!) in $\|u\|_{H^3}^2$. Apply linear Gronwall:

$$\|u(t)\|_{H^3}^2 \leq \|u_0\|_{H^3}^2\, e^{C(M)t}.$$

H^3 norm bounded on every finite interval $\Rightarrow$ no blow-up $\Rightarrow$ local theorem iterates to $[0,\infty)$.

Theorem. For every $u_0 \in H^3_{\text{per}}(-L,L)$, KdV has a unique global solution

$$u \in C([0,\infty);\, H^3_{\text{per}}(-L,L)).$$

Modern Theory

- Work in Sobolev spaces $H^s(\mathbb{R})$
- Local and global well-posedness
- Sharp results (Tao and others):
  - Global well-posedness for $s \geq -3/4$
- Conservation laws control norms

Hierarchy of Conserved Quantities

Assume

$$u, u_x, u_{xx} \to 0 \text{ as } x \to \pm\infty$$

- Mass:
$$\int u\,dx$$
- Momentum:
$$\int u^2\,dx$$
- Energy:
$$\int \left(u^3 - \frac{1}{2}u_x^2\right)dx$$
- Infinite hierarchy:

Mass Conservation

mismatch $-L, L$ $-\infty, \infty$

Goal: Show

$$\int_{-\infty}^{\infty} u\,dx = const$$

Use PDE:

$$u_t + 6uu_x + u_{xxx} = 0$$

$$\frac{d}{dt}\int_{-\infty}^{\infty} u\,dx + \int_{-\infty}^{\infty}(3u^2 + u_{xx})_x\,dx = 0$$

$$\frac{d}{dt}\int_{-\infty}^{\infty} u\,dx = 0$$

Hence, mass is conserved.

Momentum Conservation

Goal: Show

$$\int_{-\infty}^{\infty} u^2\, dx = const$$

Use PDE:

$$u_t + 6uu_x + u_{xxx} = 0$$

$$uu_t + 6u^2u_x + uu_{xxx} = 0$$

$$\frac{1}{2}(u^2)_t + (2u^3)_x + (uu_{xx} - \frac{1}{2}u_x^2)_x = 0$$

$$\frac{d}{dt}\int_{-\infty}^{\infty} \frac{1}{2}u^2\, dx + \int_{-\infty}^{\infty} (2u^3 + uu_{xx} - \frac{1}{2}u_x^2)_x\, dx = 0$$

$$\int_{-\infty}^{\infty} u^2\, dx = const$$

So, momentum is conserved.

Energy Conservation

Goal: Show

$$\int_{-\infty}^{\infty} (u^3 - \frac{1}{2}u_x^2)\, dx = const$$

$$\frac{d}{dt}\int_{-\infty}^{\infty} (u^3 - \frac{1}{2}u_x^2)\, dx = \int_{-\infty}^{\infty} (3u^2u_t - u_xu_{xt})\, dx$$

$$= \int_{-\infty}^{\infty} 3u^2u_t\, dx - (u_xu_t)\Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} u_{xx}u_t\, dx$$

$$= \int_{-\infty}^{\infty} 3u^2u_t\, dx + \int_{-\infty}^{\infty} u_{xx}u_t$$

Using PDE:

$$u_t = -6uu_x - u_{xxx}$$

$$= \int_{-\infty}^{\infty} (3u^2 + u_{xx})(-6uu_x - u_{xxx})\, dx$$

$$= \int_{-\infty}^{\infty} -(3u^2 + u_{xx})(3u^2 + u_{xx})_x\, dx$$

$$= -\int_{-\infty}^{\infty} \frac{1}{2}(3u^2 + u_{xx})_x^2\, dx$$

$$= 0$$

Hence, energy is conserved.

Infinite hierarchy

Kdv has infinite many conservational laws.

Travelling Wave Solutions

- Ansatz: $u(x,t) = f(x - ct)$
- Leads to ODE

$$u(x,t) = \frac{c}{2}\operatorname{sech}^2\left(\frac{\sqrt{c}}{2}(x - ct)\right)$$

This is a soln
not the ode.

Lax Pair

- Consider two differential operators L and P:

$$L = -\partial_x^2 - u, \quad P = -4\partial_x^3 - 6u\partial_x - 3u_x$$

- The **Lax equation** is defined as:

$$\frac{dL}{dt} = [P, L] := PL - LP$$

where $[P, L]$ is the **commutator** of P and L

- Key observation: direct computation shows

$$[P, L] = 6uu_x + u_{xxx}$$

so the Lax equation is **equivalent** to KdV:

$$\frac{dL}{dt} = [P, L] \iff u_t + 6uu_x + u_{xxx} = 0$$

- Consequence: the eigenvalue problem $L\psi = \lambda\psi$

$$-\psi_{xx} - u\psi = \lambda\psi$$

has eigenvalues λ independent of time

Inverse Scattering Transform

- A three-step algorithm to solve KdV exactly, analogous to the Fourier transform for linear PDEs

$$\begin{tikzcd}
\boxed{u(x,0)} \arrow[r, "\text{Step 1}", "\text{Direct Scattering}"'] \arrow[d, dashed, "\text{KdV (hard)}"'] & \boxed{S(0) = \{k_n, c_n(0), b(k,0)\}} \arrow[d, "\text{Step 2 Time Evolution}"] \\
\boxed{u(x,t)} & \boxed{S(t) = \{k_n, c_n(t), b(k,t)\}} \arrow[l, "\text{Step 3}"', "\text{Inverse Scattering}"]
\end{tikzcd}$$

- **Step 1** (Direct Scattering): encode $u(x,0)$ into spectral data $S(0)$
- **Step 2** (Time Evolution): k_n unchanged, $c_n(t) = c_n(0)e^{8k_n^3 t}$, $b(k,t) = b(k,0)e^{8ik^3 t}$
- **Step 3** (Inverse Scattering): recover $u(x,t)$ from $S(t)$ via GLM equation

Step 1: Direct Scattering

Goal: Encode $u(x,0)$ into scattering data via the eigenvalue problem:

$$\psi_{xx} + (\lambda + u)\psi = 0$$

Bound States ($\lambda < 0$)
Let $k_n = \sqrt{-\lambda_n} > 0$
Require $\psi \to 0$ as $x \to \pm\infty$:

$$\psi_n(x) \sim \begin{cases} e^{k_n x} & x \to -\infty \\ e^{-k_n x} & x \to +\infty \end{cases}$$

This condition yields **discrete** eigenvalues k_n and norming constants:

$$c_n(0) = \left(\int_{-\infty}^{\infty} \psi_n^2\, dx\right)^{-1}$$

Unbound States ($\lambda > 0$)
Let $k = \sqrt{\lambda} > 0$
ψ is bounded but oscillatory:

$$\psi \sim \begin{cases} e^{-ikx} + b(k,0)e^{ikx} & x \to +\infty \\ a(k,0)e^{-ikx} & x \to -\infty \end{cases}$$

- $b(k,0)$: reflection coefficient, read off from asymptotic behavior
- $a(k,0)$: transmission coefficient

looks
good.

Step 2: Time Evolution

Goal: Evolve scattering data from $\mathcal{S}(0)$ to $\mathcal{S}(t)$

Recall from Lax Pair: the second equation

$$\psi_t = P\psi, \quad P = -4\partial_x^3 - 6u\partial_x - 3u_x$$

governs how ψ evolves in time.

This determines how each piece of scattering data evolves:

Data	Comes from	Method
k_n	$L\psi = \lambda\psi$	eigenvalues conserved
$c_n(t)$	$\psi_t = P\psi$	integrate over ψ_n^2
$b(k,t)$	$\psi_t = P\psi$	asymptotic behavior as $x \to +\infty$

Key simplification: as $x \to \pm\infty$, $u \to 0$, so P reduces to $-4\partial_x^3$, giving **explicit** evolution equations

Step 2: Time Evolution

Key tool: Define auxiliary quantity R:

$$R = \psi_t - u_x\psi - 2(2\lambda - u)\psi_x$$

Properties:

1. Using KdV and $\lambda_t = 0$ (holds for both bound and unbound states):

$$\psi^2\left(\frac{R}{\psi}\right)_x = \psi R_x - \psi_x R = 0$$

so $\frac{R}{\psi}$ is a function of t only

2. Evaluate at $x \to +\infty$:

Bound states ($\lambda_n < 0$): $\psi_n \sim e^{-k_n x}$

$$\frac{R}{\psi_n} = 4\lambda_n k_n = -4k_n^3 \implies R = -4k_n^3\psi_n$$

Unbound states ($\lambda > 0$): $\psi \sim e^{-ikx} + b(k,0)e^{ikx}$

$$\frac{R}{\psi} = 4ik^3 \implies R = 4ik^3\psi$$

Step 2: Time Evolution

Goal: Derive time evolution of scattering data using $R = -4k_n^3\psi_n$ and $R = 4ik^3\psi$

Norming constants $c_n(t)$: using $R = -4k_n^3\psi_n$

$$\frac{d}{dt}(c_n^{-1}) = 2\int_{-\infty}^{\infty} \psi_n\psi_{n,t}\,dx = -8k_n^3 c_n^{-1} \implies \boxed{c_n(t) = c_n(0)e^{8k_n^3 t}}$$

Reflection coefficient $b(k,t)$: using $R = 4ik^3\psi$, evaluate at $x \to +\infty$

$$\frac{R}{\psi} = \frac{b_t e^{ikx} + 4ik^3 e^{-ikx} - 4ik^3 b e^{ikx}}{e^{-ikx} + be^{ikx}} = 4ik^3 \implies b_t = 8ik^3 b$$

$$\implies \boxed{b(k,t) = b(k,0)e^{8ik^3 t}}$$

Transmission coefficient $a(k,t)$: evaluate at $x \to -\infty$

$$\frac{R}{\psi} = \frac{a_t}{a} + 4ik^3 = 4ik^3 \implies a_t = 0 \implies \boxed{a(k,t) = a(k,0)}$$

Step 3: Inverse Scattering

Goal: Recover $u(x,t)$ from $\mathcal{S}(t)$ via the Gelfand-Levitan-Marchenko (GLM) equation:

$$K(x,y,t) + B(x+y,t) + \int_x^{\infty} K(x,z,t)B(y+z,t)\,dz = 0 \quad y > x$$

where B is defined by scattering data:

$$B(x+y,t) = \sum_{n=1}^{N} c_n(t)^2 e^{-k_n(x+y)} + \frac{1}{2\pi}\int_{-\infty}^{\infty} b(k,t)e^{ik(x+y)}dk$$

and u is recovered by:

$$u(x,t) = 2\frac{\partial}{\partial x}K(x,x,t)$$

Special case: $b(k,t) = 0$

- B reduces to: $B(x+y,t) = \sum_{n=1}^{N} c_n(t)e^{-k_n(x+y)}$
- GLM reduces to a $N \times N$ **linear algebraic system**
- **Analytical solution possible**

good.

One-Soliton Solution: $u(x,0) = 2\,\mathrm{sech}^2 x$

Step 1: Scattering equation $\psi_{xx} + (\lambda + 2\,\mathrm{sech}^2 x)\psi = 0$
Substitute $T = \tanh x \to$ Associated Legendre Equation:

$$\frac{\partial}{\partial T}\left((1-T^2)\frac{\partial\psi}{\partial T}\right) + \left(2 + \frac{\lambda}{1-T^2}\right)\psi = 0$$

Bounded solution: $\psi_1 = \frac{1}{2}\,\mathrm{sech}\, x$, giving scattering data:

$$\lambda_1 = -1, \quad k_1 = 1, \quad c_1(0) = 2, \quad b(k,0) = 0$$

Step 2: Time evolution:

$$c_1(t) = 2e^{8t}, \quad b(k,t) = 0$$

Step 3: $b = 0$, so $B(x+y,t) = 2e^{8t-x-y}$
Set $K(x,y,t) = w(x,t)e^{-y}$, GLM reduces to:

$$w(1 + e^{8t-2x}) = -2e^{8t-x} \implies K(x,x,t) = \frac{-2e^{8t-2x}}{1+e^{8t-2x}}$$

Recover u:

$$u(x,t) = 2\frac{\partial}{\partial x}K(x,x,t) = \boxed{2\,\mathrm{sech}^2(x-4t)}$$

Two-Soliton Solution: $u(x,0) = 6\,\mathrm{sech}^2 x$

Step 1: Scattering equation $\psi_{xx} + (\lambda + 6\,\mathrm{sech}^2 x)\psi = 0$
Two bounded solutions, giving scattering data:

$$\psi_1 = \tfrac{1}{4}\,\mathrm{sech}^2 x, \quad k_1 = 2, \quad c_1(0) = 12,$$

$$\psi_2 = \tfrac{1}{2}\tanh x\,\mathrm{sech}\, x, \quad k_2 = 1, \quad c_2(0) = 6, \quad b(k,0) = 0$$

Step 2: Time evolution:

$$c_1(t) = 12e^{64t}, \quad c_2(t) = 6e^{8t}, \quad b(k,t) = 0$$

Step 3: $B(x+y,t) = 12e^{64t}e^{-2(x+y)} + 6e^{8t}e^{-(x+y)}$

Set $K(x,y,t) = w_1(x,t)e^{-2y} + w_2(x,t)e^{-y}$, GLM reduces to 2×2 linear system:

$$\begin{pmatrix} 1 + \frac{12e^{64t-4x}}{4} & \frac{12e^{64t-3x}}{3} \\ \frac{6e^{8t-3x}}{3} & 1 + \frac{6e^{8t-2x}}{2} \end{pmatrix}\begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = -\begin{pmatrix} 12e^{64t-2x} \\ 6e^{8t-x} \end{pmatrix}$$

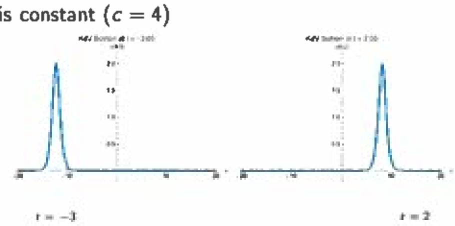
Solving and recovering u:

$$u(x,t) = 2\frac{\partial}{\partial x}K(x,x,t) = \boxed{12\,\frac{3 + 4\cosh(2x-8t) + \cosh(4x-64t)}{(3\cosh(x-28t) + \cosh(3x-36t))^2}}$$

could connect with above theory.

One-Soliton Evolution

- Initial condition: $u(x,0) = 2\,\mathrm{sech}^2 x$
- One-Soliton Solution: $u(x,t) = 2\,\mathrm{sech}^2(x-4t)$
- Single soliton travels without changing shape
- Speed is constant ($c = 4$)



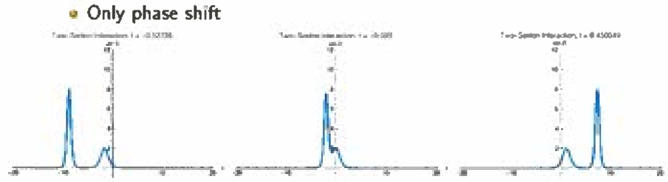
$t = -3$ $t = 2$

Soliton profiles at different times

could draw track pattern in t, x

Two-Soliton Interaction

- Initial condition: $u(x,0) = 6\,\mathrm{sech}^2 x$
- Two-Soliton Solution: $u(x,t) = \frac{12(3+4\cosh(2x-8t)+\cosh(4x-64t))}{(3\cosh(x-28t)+\cosh(3x-36t))^2}$
- Two solitons with different speeds
- Interaction is elastic:
  - No change in shape
  - Only phase shift



Soliton profiles at different times

Comparison with Burgers Equation

$$u_t + uu_x = 0$$

- Burgers: shock formation
- KdV: dispersion prevents shocks
- Balance yields stable structures

Slowly Varying Depth

- Let $h = h(x)$ vary slowly
- Leads to variable coefficient KdV
- Effects:
  - Soliton amplitude changes
  - Speed adapts to depth

Summary

- KdV models shallow water waves
- Rich mathematical structure
- Solitons and integrability
- Robust under perturbations

Sections

- Modeling
- Interpretation of Terms: Steepening and Dispersion
- Basic Existence and Uniqueness

  Jiajia Wang
- Sobolev Regularity Theory

  Sneha Poudel
- Conserved Quantities

  Quyen Nguyen
- Split-Step Fourier Schemes on Periodic Domain $-L < x < L$

  Allan Newman
- Aliasing Effects in Fourier Schemes
- Lax Pair Formulation
- Inverse Scattering Formulation

  Qiuyun Jin
- Single Eigenvalue Soliton